

TOULOUSE BLAGNAC AP, France

WMO: 076300

Lat: 43.6211N

Lon: 1.3789E

Elev: 499

StdP: 14.43

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	25.3	28.4	19.2	14.7	30.9	22.5	17.2	32.6	27.6	50.8	24.6	49.1	5.5	280	0.510

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.2	92.0	69.4	88.6	68.5	85.7	67.6	72.2	85.8	70.8	83.8	69.4	81.8	9.0	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.0	104.8	77.5	66.5	99.2	76.5	65.1	94.6	75.1	36.1	85.6	34.9	84.1	33.7	81.9	84.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
24.2	21.0	18.8	DB	21.6	97.5	3.8	3.7	18.8	100.2	16.6	102.4	14.5	104.4	11.8	107.1	(4)
			WB	20.7	74.9	3.9	2.3	17.9	76.5	15.6	77.9	13.5	79.2	10.6	80.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	57.2	43.2	44.6	50.2	54.5	61.0	68.4	72.2	72.3	66.3	59.9	49.3	44.2	(6)
		DBStd	12.20	6.53	6.62	5.62	5.50	6.01	6.55	5.92	5.87	5.72	6.50	6.67	7.02	(7)
		HDD50	773	222	168	66	19	3	0	0	0	7	90	198		(8)
		HDD65	3581	675	572	458	316	153	39	13	9	51	181	471	643	(9)
		CDD50	3418	12	16	73	154	343	551	687	691	489	313	69	20	(10)
		CDD65	751	0	0	0	1	27	140	235	235	91	22	0	0	(11)
		CDH74	6757	0	0	1	35	259	1306	2213	2142	718	83	0	0	(12)
		CDH80	2440	0	0	0	2	48	460	869	864	190	7	0	0	(13)
(14)	Wind	WSAvg	8.7	8.6	9.3	9.8	9.7	9.3	8.7	8.5	7.9	7.8	8.3	8.2	8.0	(14)
(15)	Precipitation	PrecAvg	25.7	2.1	1.9	2.0	2.6	2.9	2.4	1.7	1.9	2.0	2.1	2.1	2.0	(15)
(16)		PrecMax	36.2	5.7	3.7	4.6	6.7	7.1	10.4	7.3	5.9	5.9	6.4	5.7	6.6	(16)
(17)		PrecMin	16.0	0.1	0.2	0.0	0.4	0.8	0.3	0.1	0.4	0.0	0.4	0.1	0.1	(17)
(18)		PrecStd	4.8	1.2	1.0	1.1	1.4	1.3	1.6	1.2	1.1	1.4	1.5	1.1	1.3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	60.5	65.0	72.2	78.6	85.0	94.0	95.7	98.4	89.0	80.5	68.7	61.5		(19)
(20)		0.4% MCWB	53.8	53.4	55.5	61.5	64.9	70.0	69.4	70.1	67.2	64.4	59.4	55.2		(20)
(21)		2% DB	57.1	60.1	67.2	73.1	80.4	88.6	91.4	92.6	85.2	76.0	64.9	58.5		(21)
(22)		2% MCWB	52.2	51.4	54.4	58.0	63.3	68.3	69.2	69.6	65.8	63.9	57.8	53.4		(22)
(23)		5% DB	55.0	57.0	63.5	69.2	76.6	85.2	88.3	88.3	81.7	72.8	62.0	56.3		(23)
(24)		5% MCWB	51.0	50.2	52.6	56.2	62.4	67.1	68.5	69.1	64.8	62.3	56.5	52.2		(24)
(25)		10% DB	52.8	54.3	60.2	65.5	72.8	81.8	85.1	84.7	78.1	69.9	59.3	54.0		(25)
(26)		10% MCWB	49.4	49.1	51.6	54.6	60.4	66.0	67.7	67.9	63.8	61.6	54.8	50.7		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	55.3	54.7	58.1	62.2	67.8	73.3	73.5	74.4	70.6	67.3	61.0	56.0		(27)
(28)		0.4% MCDB	58.4	60.8	66.9	75.1	80.6	88.4	88.5	87.8	83.5	75.5	66.8	59.9		(28)
(29)		2% WB	53.0	52.8	55.6	59.6	65.3	70.7	71.9	72.4	68.7	65.6	59.0	54.4		(29)
(30)		2% MCDB	56.3	58.2	63.4	70.5	76.8	84.8	85.9	85.3	79.5	72.2	63.5	57.6		(30)
(31)		5% WB	51.5	51.2	54.0	57.6	63.3	68.7	70.5	70.9	67.2	64.3	57.1	52.7		(31)
(32)		5% MCDB	54.5	55.7	61.4	66.6	73.5	81.7	83.7	83.3	77.1	70.3	61.2	55.8		(32)
(33)		10% WB	49.7	49.7	52.5	55.7	61.5	67.1	69.1	69.5	65.5	62.8	55.3	51.0		(33)
(34)		10% MCDB	52.4	53.6	58.9	63.3	70.5	78.9	81.4	81.5	74.9	68.5	58.7	53.9		(34)
(35)	Mean Daily Temperature Range	MDBR	11.9	14.1	16.4	17.2	18.0	19.7	20.2	20.2	19.3	16.1	12.9	12.2		(35)
(36)		5% DB MCDBR	14.0	18.6	23.6	24.3	24.5	25.7	26.6	26.7	25.0	19.4	16.4	14.4		(36)
(37)		5% DB MCWBR	9.9	11.4	12.7	12.0	11.4	9.8	9.2	9.1	9.5	9.3	10.2	10.2		(37)
(38)		5% WB MCDBR	12.2	16.1	18.5	20.3	21.1	22.7	22.4	22.3	20.4	15.3	13.9	12.8		(38)
(39)		5% WB MCWBR	9.2	10.7	10.6	10.9	10.9	9.8	9.1	9.2	8.7	7.9	9.2	9.5		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.321	0.336	0.374	0.398	0.407	0.418	0.410	0.410	0.379	0.365	0.339	0.319		(40)
(41)		taud	2.476	2.412	2.310	2.249	2.281	2.284	2.306	2.317	2.388	2.432	2.460	2.480		(41)
(42)		Ebn at Noon	257	271	273	275	275	271	272	268	268	257	246	246		(42)
(43)		Edh at Noon	23	29	36	42	41	41	40	38	33	28	23	21		(43)
(44)	All-Sky Solar Radiation	RadAvg	457	756	1147	1485	1695	1947	1929	1722	1379	916	540	425		(44)
(45)		RadStd	41	84	107	158	124	138	136	104	70	74	59	56		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (24 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page